

Practice Exam Set B — Solutions

Part I: Generative Models

Exercise A — VAE with MoG prior

Answer: A

With $z' = 2$, $\mathbf{x}' = [2, -2]^\top$, $q(z|\mathbf{x}') = \mathcal{N}(z | x_1 - x_2/2, 1) = \mathcal{N}(z | 3, 1)$:

Term 1 — $\ln p(\mathbf{x}'|z' = 2)$: Mean $= [2, -2]^\top$, $\|\mathbf{x}' - \text{mean}\|^2 = 0$.

$$\ln p(\mathbf{x}'|z') = \ln \mathcal{N}([2, -2]^\top | [2, -2]^\top, \mathbf{I}_2) = 0 - \ln(2\pi) = -\ln(2\pi).$$

Term 2 — $\ln p(z' = 2)$: Both components evaluate to e^{-2} (symmetric about $z = 2$):

$$p(2) = \frac{1}{2}\mathcal{N}(2|0, 1) + \frac{1}{2}\mathcal{N}(2|4, 1) = \frac{1}{2} \cdot \frac{e^{-2}}{\sqrt{2\pi}} + \frac{1}{2} \cdot \frac{e^{-2}}{\sqrt{2\pi}} = \frac{e^{-2}}{\sqrt{2\pi}}, \quad \ln p(2) = -2 - \frac{1}{2} \ln(2\pi).$$

Term 3 — $\ln q(z' = 2|\mathbf{x}')$: $q = \mathcal{N}(z | 3, 1)$:

$$\ln q(2|\mathbf{x}') = -\frac{(2-3)^2}{2} - \frac{1}{2} \ln(2\pi) = -\frac{1}{2} - \frac{1}{2} \ln(2\pi).$$

ELBO:

$$\begin{aligned} \text{ELBO} &\approx -\ln(2\pi) + \left(-2 - \frac{1}{2} \ln(2\pi)\right) - \left(-\frac{1}{2} - \frac{1}{2} \ln(2\pi)\right) \\ &= -\ln(2\pi) - 2 - \frac{1}{2} \ln(2\pi) + \frac{1}{2} + \frac{1}{2} \ln(2\pi) = \boxed{-\ln(2\pi) - \frac{3}{2}}. \end{aligned}$$

Exercise B — Normalizing flows (affine coupling)

B.1

Answer: C

Inverting $T(\mathbf{u}) = [u_1, e^{s(u_1)}u_2 + t(u_1)]^\top$ gives $u_1 = z_1$ and $u_2 = (z_2 - t(z_1))e^{-s(z_1)}$, which is valid for *all* C^1 functions s, t since $e^{-s} > 0$ always.

B.2

Answer: B

J_T is lower-triangular: $J_T = \begin{bmatrix} 1 & 0 \\ \cdot & e^{s(u_1)} \end{bmatrix}$, so $\det J_T = e^{s(u_1)}$.

B.3

Answer: A

With $s = 0$, $t(u_1) = u_1^2$: $T(\mathbf{u}) = [u_1, u_2 + u_1^2]^\top$. $T^{-1}([1, 2]^\top) = [1, 2 - 1^2]^\top = [1, 1]^\top$. $\det J_T = e^0 = 1$.

$$p_{\mathbf{x}}([1, 2]^\top) = \frac{p_{\mathbf{u}}([1, 1]^\top)}{1} = \frac{1}{2\pi} e^{-(1+1)/2} = \boxed{\frac{e^{-1}}{2\pi}}.$$

Exercise C — DDPM**Answer: C**

The forward process q is a Markov chain: $q(\mathbf{x}_t|\mathbf{x}_{t-1}, \dots) = q(\mathbf{x}_t|\mathbf{x}_{t-1})$. In the chain $\mathbf{x}_1 \rightarrow \mathbf{x}_2 \rightarrow \mathbf{x}_3 \rightarrow \mathbf{x}_4 \rightarrow \mathbf{x}_5$, conditioning on $\mathbf{x}_4$ blocks the path from $\mathbf{x}_3$ to $\mathbf{x}_5$ (d-separation), so $\mathbf{x}_5 \perp \mathbf{x}_3|\mathbf{x}_4$. Therefore:

$$q(\mathbf{x}_3|\mathbf{x}_4, \mathbf{x}_1, \mathbf{x}_5) = q(\mathbf{x}_3|\mathbf{x}_4, \mathbf{x}_1).$$

The other options fail: (A) ignores the informative $\mathbf{x}_4$; (B) wrong conditioning set; (D) $\mathbf{x}_5$ is redundant given $\mathbf{x}_4$ but $\mathbf{x}_1$ is not.

Part II: Representation Learning

Exercise D — Pull-back metric (tanh)

Answer: A

For $f(\mathbf{x}) = W_2 \tanh(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$, by the chain rule:

$$J_{\mathbf{x}} = W_2 \underbrace{\text{diag}(1 - \tanh^2(W_1 \mathbf{x} + \mathbf{b}_1))}_{\Lambda} W_1.$$

$$G_{\mathbf{x}} = J_{\mathbf{x}}^{\top} J_{\mathbf{x}} = (W_2 \Lambda W_1)^{\top} (W_2 \Lambda W_1) = W_1^{\top} \Lambda W_2^{\top} W_2 \Lambda W_1.$$

(Λ is symmetric since it is diagonal.)

Exercise E — Curve energy

E.1: For $f(\mathbf{x}) = [2x_1^2, x_1^2 + 3x_2^2, 2x_1 + x_2^2]^{\top}$:

$$J_{\mathbf{x}} = \begin{bmatrix} 4x_1 & 0 \\ 2x_1 & 6x_2 \\ 2 & 2x_2 \end{bmatrix}.$$

$$G_{\mathbf{x}} = J^{\top} J = \begin{bmatrix} (4x_1)^2 + (2x_1)^2 + 4 & 0 + 12x_1x_2 + 4x_2 \\ 0 + (6x_2)^2 + (2x_2)^2 & \end{bmatrix} = \begin{bmatrix} 20x_1^2 + 4 & 4x_2(3x_1 + 1) \\ 4x_2(3x_1 + 1) & 40x_2^2 \end{bmatrix}.$$

E.2: For $\mathbf{c}_1(t) = [t, 0]^{\top}$: $G_{\mathbf{c}_1(t)} = \begin{bmatrix} 20t^2 + 4 & 0 \\ 0 & 0 \end{bmatrix}$, $\dot{\mathbf{c}}_1^{\top} G \dot{\mathbf{c}}_1 = 20t^2 + 4$:

$$E(\mathbf{c}_1) = \int_0^1 (20t^2 + 4) dt = \frac{20}{3} + 4 = \boxed{\frac{32}{3}}.$$

For $\mathbf{c}_2(t) = [0, t]^{\top}$: $G_{\mathbf{c}_2(t)} = \begin{bmatrix} 4 & 0 \\ 0 & 40t^2 \end{bmatrix}$, $\dot{\mathbf{c}}_2^{\top} G \dot{\mathbf{c}}_2 = 40t^2$:

$$E(\mathbf{c}_2) = \int_0^1 40t^2 dt = \boxed{\frac{40}{3}}.$$

Exercise F — Bregman divergence

For $F(p) = \sum_i p_i \ln p_i$: $\nabla F(q)_i = \frac{\partial}{\partial q_i} \sum_j q_j \ln q_j = 1 + \ln q_i$.

$$\begin{aligned} D_F(p, q) &= \sum_i p_i \ln p_i - \sum_i q_i \ln q_i - \sum_i (1 + \ln q_i)(p_i - q_i) \\ &= \sum_i p_i \ln p_i - \sum_i q_i \ln q_i - \underbrace{\left(\sum_i p_i - \sum_i q_i \right)}_{=0} - \sum_i p_i \ln q_i + \sum_i q_i \ln q_i \\ &= \sum_i p_i \ln p_i - \sum_i p_i \ln q_i = \sum_i p_i \ln \frac{p_i}{q_i} = \text{KL}(p \| q). \quad \square \end{aligned}$$

Part III: Graph Models

Exercise G — Graph metrics (path $a-b-c-d-e$)

G.1: For a path graph, no two neighbours of any node are themselves connected, so all clustering coefficients are 0:

$$c_b = 0 \quad (\mathcal{N}(b)=\{a,c\}, \text{ but no edge } a-c), \quad c_a = 0 \quad (d_a = 1, \text{ convention}).$$

G.2: From $e_a = (1/\lambda)e_b$ and the given $e_a = 0.29$, $e_b = 0.50$:

$$\lambda = \frac{e_b}{e_a} = \frac{0.50}{0.29} \approx 1.72 \approx \sqrt{3}.$$

Node c has $\mathcal{N}(c) = \{b, d\}$, and $e_b = e_d = 0.50$:

$$e_c = \frac{1}{\lambda}(e_b + e_d) = \frac{1.00}{1.72} \approx \boxed{0.58}.$$

(Exact: $e_c = 1/\sqrt{3}$.)

Exercise H — Graph Fourier transform and convolution

Path $a-b-c$, filter $h[k] = 1$ for $0 \leq k \leq 2$.

H.1: Compute A and A^2 : $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $A^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.

$\mathbf{y} = (I + A + A^2)\mathbf{x}$ with $\mathbf{x} = [1, 0, 1]^\top$:

$$I + A + A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2+0+1 \\ 1+0+1 \\ 1+0+2 \end{bmatrix} = \boxed{[3, 2, 3]^\top}.$$

H.2: $M = \sum_{k=0}^2 h[k]\Lambda^k = I + \Lambda + \Lambda^2$ with $\Lambda = \text{diag}(-\sqrt{2}, 0, \sqrt{2})$:

$$M = \text{diag}(1 + (-\sqrt{2}) + 2, 1 + 0 + 0, 1 + \sqrt{2} + 2) = \boxed{\text{diag}(3 - \sqrt{2}, 1, 3 + \sqrt{2})}.$$

Exercise I — WL test (4-cycle with chord)

4-cycle $a-b-c-d-a$ plus edge $b-d$. Degrees: $d_a = d_c = 2$, $d_b = d_d = 3$.

I.1:

- **A,B TRUE:** $\ell_a^{(0)} = \ell_c^{(0)} = 2$ and $\ell_b^{(0)} = \ell_d^{(0)} = 3$.
- **C TRUE:** $\mathcal{N}(a) = \{b, d\}$ and $\mathcal{N}(c) = \{b, d\} \Rightarrow$ same multiset $\{3, 3\}$.
- **D TRUE:** $\mathcal{N}(b) = \{a, c, d\}$ and $\mathcal{N}(d) = \{a, b, c\} \Rightarrow$ same multiset $\{2, 2, 3\}$.
- **E FALSE:** $\ell_a^{(1)} = \text{hash}(\{3, 3\}) \neq \text{hash}(\{2, 2, 3\}) = \ell_b^{(1)}$.

True: A, B, C, D

I.2: Round 0 partition: $\{\{a, c\}, \{b, d\}\}$. After round 1 the partition is unchanged (same multisets).

1 round

Exercise J — Message passing (4-cycle, asymmetric features)

$w_0 = 0$, $w_1 = w_2 = 1$, $h^{(0)} = [1, 2, 1, 2]^\top$ for (a, b, c, d) .

J.1: $\mathcal{N}(a) = \{b, d\}$, $\mathcal{N}(b) = \{a, c\}$, $\mathcal{N}(c) = \{b, d\}$, $\mathcal{N}(d) = \{c, a\}$:

$$\begin{aligned} h_a^{(1)} &= 0 + 1 \cdot 1 + 1 \cdot (2 + 2) = 5, & h_b^{(1)} &= 0 + 1 \cdot 2 + 1 \cdot (1 + 1) = 4, \\ h_c^{(1)} &= 0 + 1 \cdot 1 + 1 \cdot (2 + 2) = 5, & h_d^{(1)} &= 0 + 1 \cdot 2 + 1 \cdot (1 + 1) = 4. \end{aligned}$$

J.2: $M = I + A$ where A is the 4-cycle adjacency (ordering a, b, c, d):

$$M = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

J.3: For identity $h^{(1)} = h^{(0)}$: we need $w_0 = 0$, $w_1 = 1$, $w_2 = 0$.